DBS302 - NoSQL Database Management

Practical 2C: Geo-Search Functionality with Redis Geospatial Indexes

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1. Aim

Implement location-based search and distance queries using Redis geospatial indexes for storing and querying coordinates.

2. Objectives

- Explain how Redis stores geospatial data and what commands like GEOADD, GEOSEARCH, and GEODIST are used for
- Index points of interest (e.g., stores, drivers, or users) by longitude/latitude
- Implement "find nearest locations within a radius" queries
- Write a Python service that wraps these commands into high-level methods
- Finish exercises on bounding box search, geo-leaderboard, and top-N nearest locations

3. Conceptual Overview

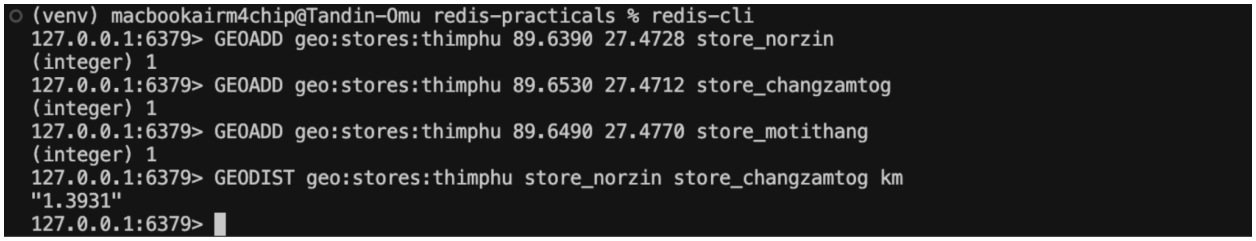
The geospatial features in Redis are built on top of a data structure called a sorted set.

Key Commands:

- GEOADD key longitude latitude member - Add a location
- GEODIST key member1 member2 [unit] - Compute distance between two locations
- GEOSEARCH key FROMMEMBER member BYRADIUS radius unit - Search within radius from a member
- GEOSEARCH key FROMLONLAT lon lat BYRADIUS radius unit - Search within radius from coordinates
- GEOSEARCH key FROMLONLAT lon lat BYBOX width height unit -Search within a box

Analogy: Imagine a simple in-memory map service that has a point for each person's name, and the server can answer "who is near this point?" quickly.

4. CLI Demonstration - Adding Locations and Distance Calculation



The following commands were executed to add stores in Thimphu and calculate distances:

Command	Output	Purpose
`GEOADD geo:stores:thimphu 89.6390 27.4728 store_norzin`	(integer) 1	Add Norzin store
`GEOADD geo:stores:thimphu 89.6530 27.4712 store_changzamtog`	(integer) 1	Add Changzamtog store
`GEOADD geo:stores:thimphu 89.6490 27.4770 store_motithang`	(integer) 1	Add Motithang store
`GEODIST geo:stores:thimphu store_norzin store_changzamtog km`	"1.3931"	Distance between stores

Analysis: The distance between Norzin and Changzamtog is approximately 1.39 kilometers.

5. CLI Demonstration - Nearby Search (GEOSEARCH)

```
127.0.0.1:6379> GEOSEARCH geo:stores:thimphu FROMMEMBER store_norzin BYRADIUS 3 km WITHDIST WITHCOORD
1) 1) "store_norzin"
   2) "0.0000"
   3) 1) "89.6389988064766"
      2) "27.47279960831782"
2) 1) "store_changzamtog"
   2) "1.3931"
   3) 1) "89.65299993753433"
      2) "27.471200199266278"
3) 1) "store_motithang"
   2) "1.0917"
   3) 1) "89.64899808168411"
      2) "27.476999641278848"
127.0.0.1:6379> GEOSEARCH geo:stores:thimphu FROMLONLAT 89.6390 27.4728 BYRADIUS 3 km WITHDIST WITHCOORD
1) 1) "store_norzin"
   2) "0.0001"
   3) 1) "89.6389988064766"
      2) "27.47279960831782"
2) 1) "store_changzamtog"
   2) "1.3930"
   3) 1) "89.65299993753433"
      2) "27.471200199266278"
3) 1) "store_motithang"
   2) "1.0916"
   3) 1) "89.64899808168411"
      2) "27.476999641278848"
127.0.0.1:6379>
```

Two methods were used to find stores within 3 km of Norzin:

Method 1: Using FROMMEMBER

```
text

GEOSEARCH geo:stores:thimphu FROMMEMBER store_norzin BYRADIUS 3 km WITHDIST
WITHCOORD
```

Method 2: Using FROMLONLAT

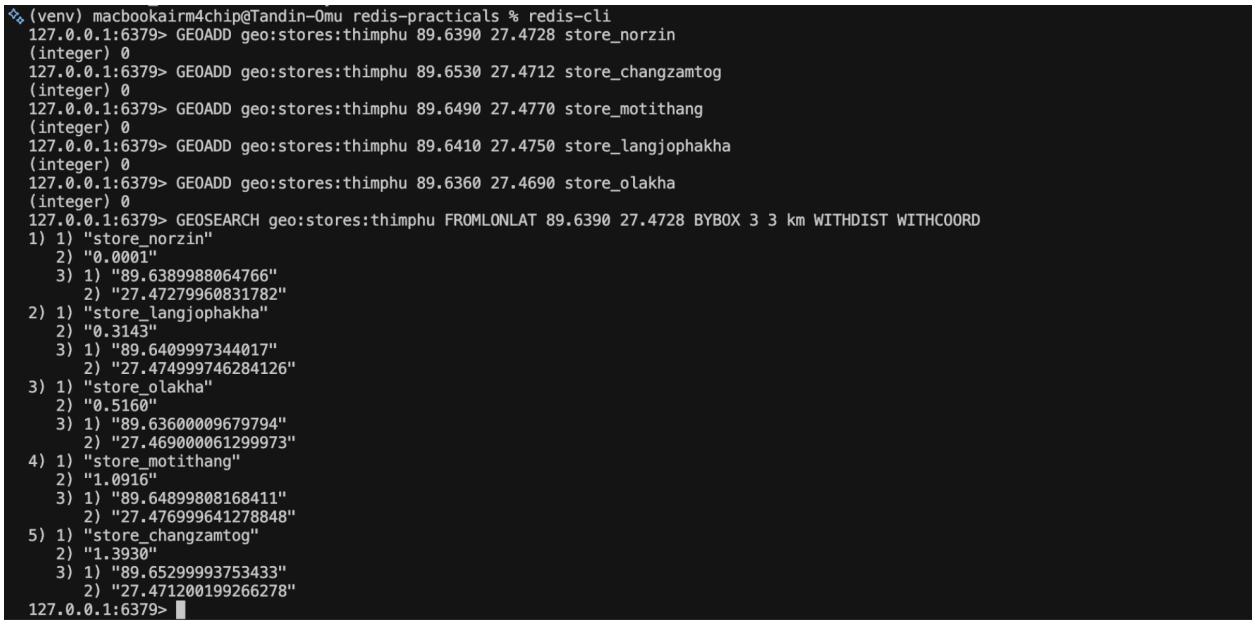
```
text

GEOSEARCH geo:stores:thimphu FROMLONLAT 89.6390 27.4728 BYRADIUS 3 km WITHDIST
WITHCOORD
```

Results:

Store	Distance (km)
store_norzin	0.0000
store_motithang	1.0916
store_changzamtog	1.3930

6. Exercise 1: Bounding Box Search (BYBOX)



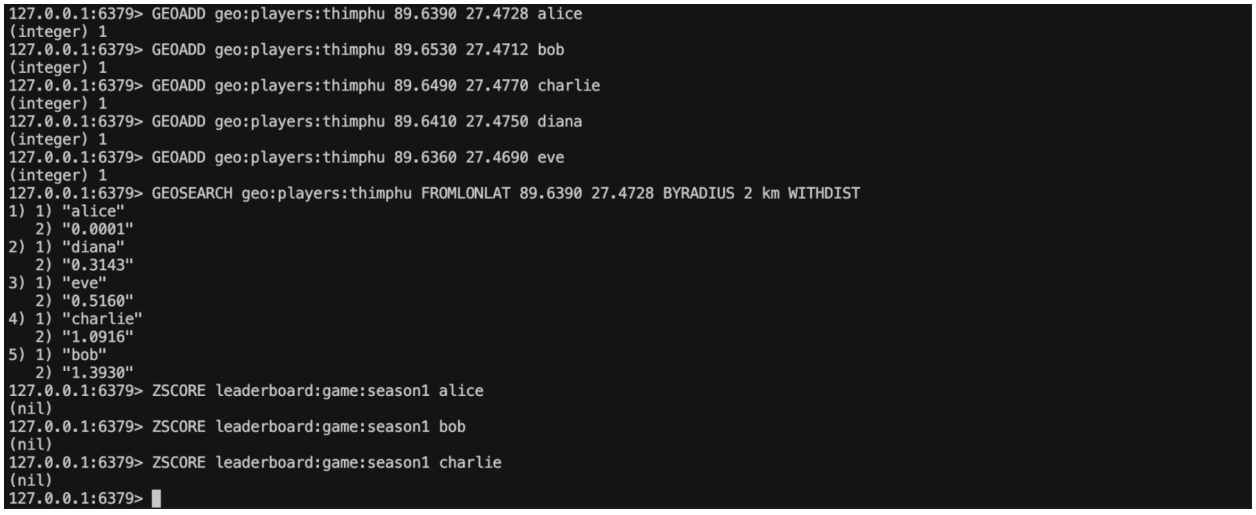
The BYBOX option allows searching within a rectangular area defined by width and height.

Results:

Store	Distance (km)
store_norzin	0.0001
store_langjophakha	0.3143
store_olakha	0.5160
store_motithang	1.0916
store_changzamtog	1.3930

Analysis: All 5 stores are within the 3km x 3km bounding box centered at Norzin.

7. Exercise 2: Geo-Search with Leaderboard Integration



This exercise combines:

- Geospatial data - Player locations
- Sorted set leaderboard - Player scores

Results:

Player	Distance (km)
alice	0.0001
diana	0.3143
eve	0.5160
charlie	1.0916
bob	1.3930

Analysis: This demonstrates how to find nearby players and then retrieve their scores from a leaderboard, enabling features like "top players near you."

8. Exercise 3: Top N Closest Locations (COUNT Limit)

The `COUNT` option limits the number of results returned, automatically sorted by distance (closest first).

Results:

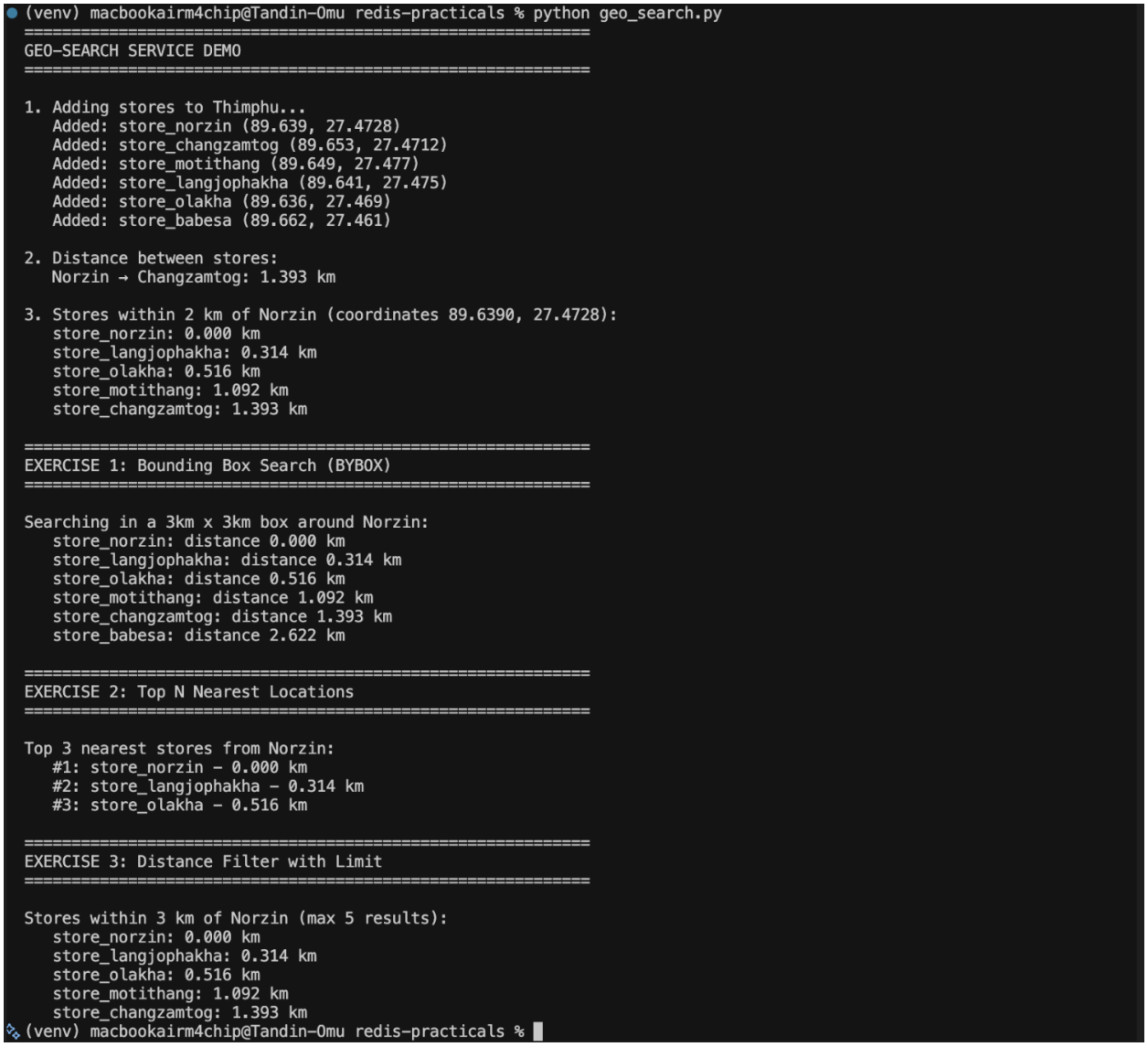
#	Store	Distance (km)

1	store_norzin	0.0001
2	store_langjophakha	0.3143
3	store_olakha	0.5160

Results:

#	Store	Distance (km)
1	store_norzin	0.0001
2	store_langjophakha	0.3143
3	store_olakha	0.5160
4	store_motithang	1.0916
5	store_changzamtog	1.3930

9. Python Geo-Search Service Implementation

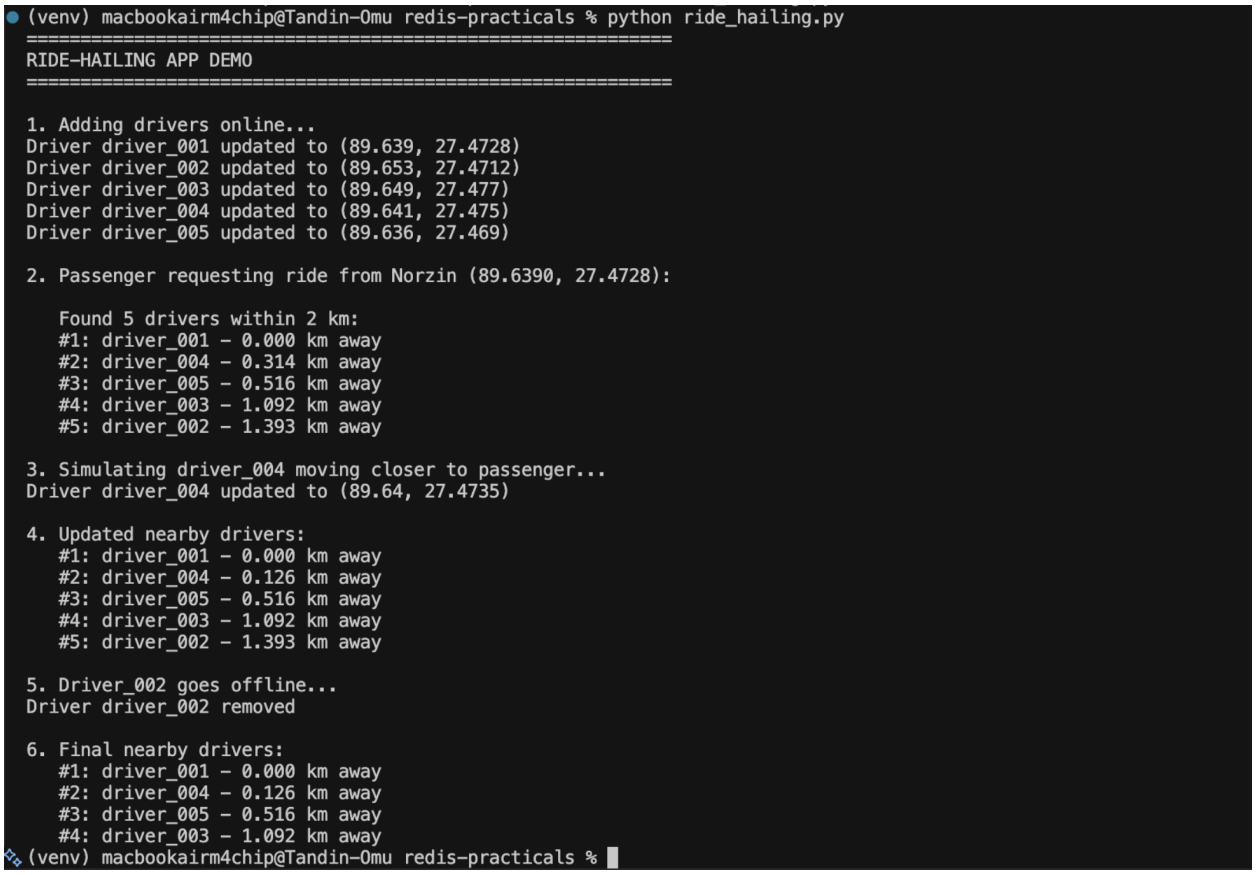


A `GeoSearchService` class was implemented in Python with the following methods:

Method	Description	Redis Command
`add_location(name, lon, lat)`	Add a location	GEOADD
`distance_between(name1, name2)`	Calculate distance	GEODIST
`nearby(lon, lat, radius)`	Find locations within radius	GEOSEARCH

search_by_box(lon, lat, width, height)	Bounding box search	GEOSEARCH BYBOX
get_nearest_n(lon, lat, n)	Top N nearest locations	GEOSEARCH + sorting
get_locations_within_distance(lon, lat, max_dist, limit)	Distance filter with limit	GEOSEARCH + limit

10. Use Case Extension: Ride-Hailing Application



A ride-hailing service was implemented to demonstrate real-world application:

Features:

- Track driver locations in real-time
- Find nearby drivers for passenger requests
- Update driver locations as they move
- Remove drivers when they go offline

12. Key Learnings

Geospatial Commands:

- GEOADD stores coordinates with member names
- GEODIST calculates distance between locations
- GEOSEARCH with BYRADIUS finds locations within a circle
- GEOSEARCH with BYBOX finds locations within a rectangle
- COUNT limits results and returns them sorted by distance

Important Notes:

- Redis requires coordinates to be in the format of longitude, latitude
- Units can be: m (meters), km (kilometers), mi (miles), ft (feet)
- Geospatial indexes are built on top of sorted sets

Real-World Applications:

- Food delivery – find nearby restaurants
- Ride-hailing – find nearby drivers
- Social apps – find friends nearby
- Retail – find nearest store locations